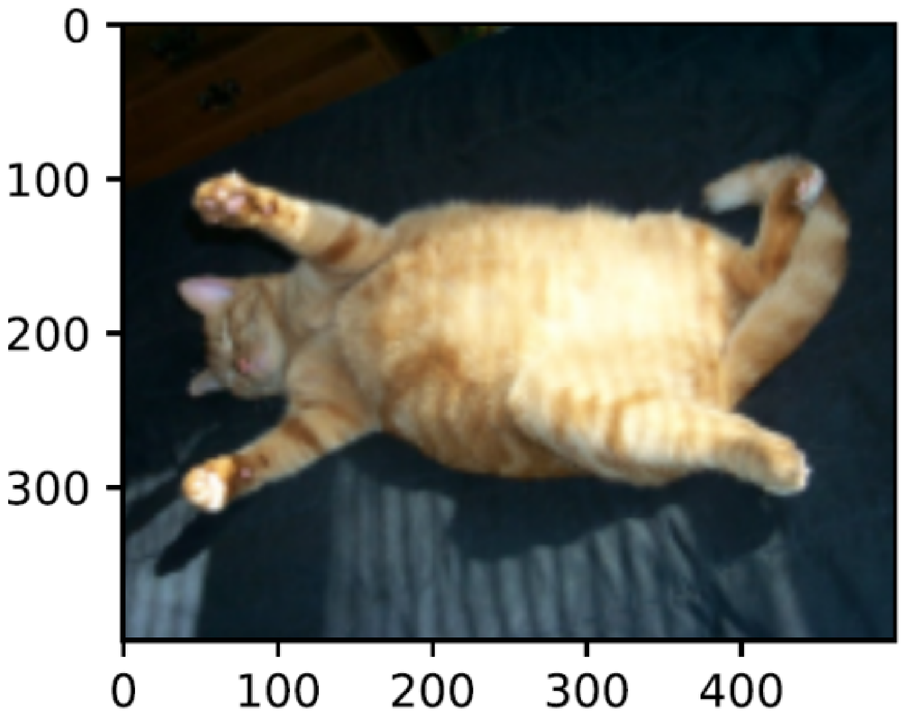
6.1 Image augmentation

- in Section 3.7, we mentioned that large datasets are a prerequisite for the success of deep neural networks in various applications
- *image augmentation* generates similar but distinct training examples after a series of random changes to the training images, thereby expanding the size of the training set
- alternatively, image augmentation can be motivated by the fact that random tweaks of training examples allow models to rely less on certain attributes, thereby improving their generalization ability
- for example, we can crop an image in different ways to make the object of interest appear in different positions, thereby reducing the dependence of a model on the position of the object
- we can also adjust factors such as brightness and color, to reduce a model's sensitivity to color
- it is probably true that image augmentation was indispensable for the success of AlexNet, at that time
- in this section, we will discuss this widely used technique in computer vision

6.1 Image augmentation

- in our investigation of common image augmentation methods, we will use the following 400×500 image as an example
- most image augmentation methods have a certain degree of randomness
- thus, to make it easier for us to observe the effect of image augmentation, we run the image augmentation method multiple times on the input image, and show all the results



6.1 Image augmentation

- *flipping* the image left and right usually does not change the category of the object
- this is one of the earliest and most widely used methods of image augmentation
- next, we flip the image left and right with a 50% chance



6.1 Image augmentation

- flipping up and down is not as common as flipping left and right
- but, at least for this example image, flipping up and down does not hinder recognition
- next, we flip the image up and down with a 50% chance



6.1 Image augmentation

- in the example image we used, the cat is in the middle of the image, but this may not be the case, in general
- in Section 3.5, we explained that the pooling layer can reduce the sensitivity of a convolutional layer to the target position
- in addition, we can also randomly *crop* the image to make objects appear in different positions in the image at different scales, which can also reduce the sensitivity of a model to the target position

6.1 Image augmentation

- in the example below, we randomly crop an area with an area of 10% $\sim$ 100% of the original area each time, and the ratio of width to height of this area is randomly selected from 0.5 $\sim$ 2
- then, the width and height of the region are both scaled to 200 pixels
- unless otherwise specified, the random number between a and b in this section refers to a continuous value obtained by random and uniform sampling from the interval $[a, b]$



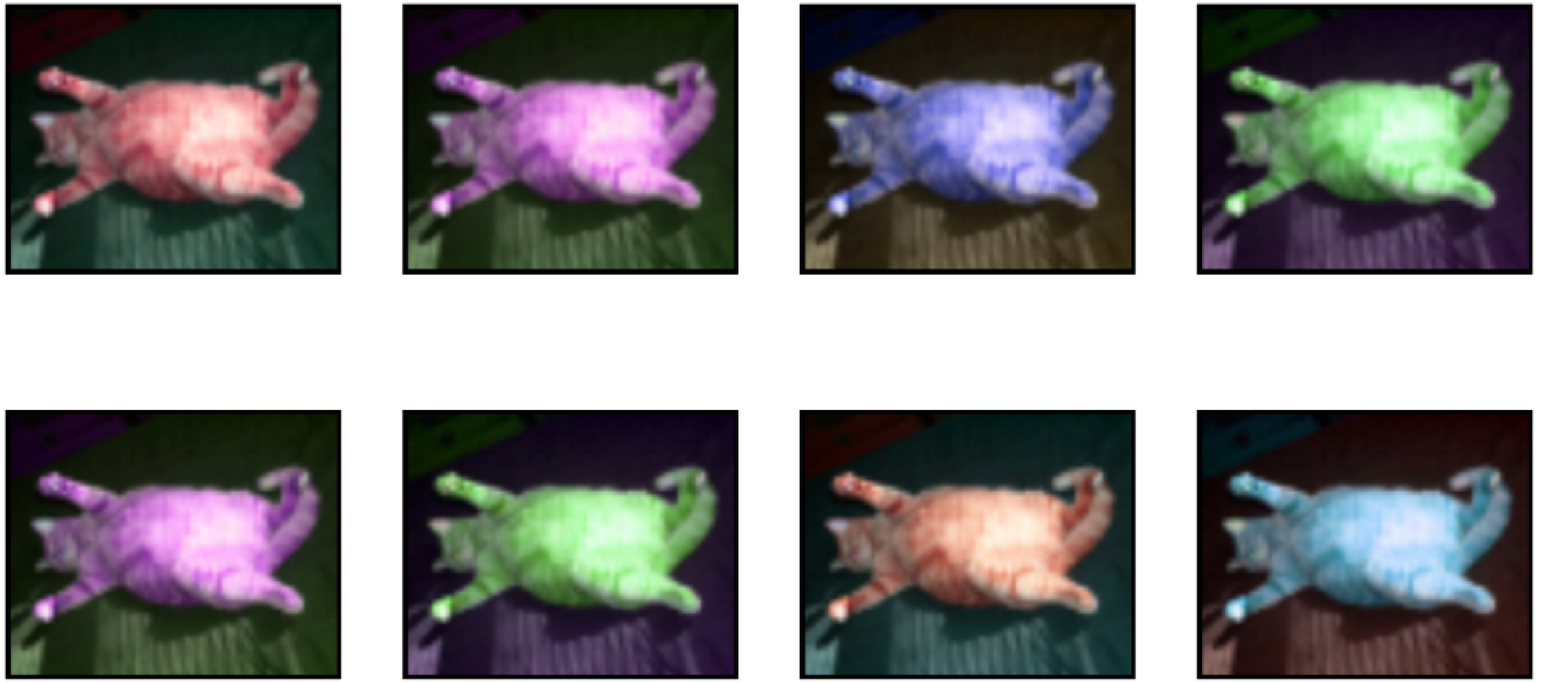
6.1 Image augmentation

- another augmentation method is *changing colors*
- we can change four aspects of the image color: brightness, contrast, saturation, and hue
- in the example below, we randomly change the brightness of the image to a value between 50% $(1 - 0.5)$ and 150% $(1 + 0.5)$ of the original image



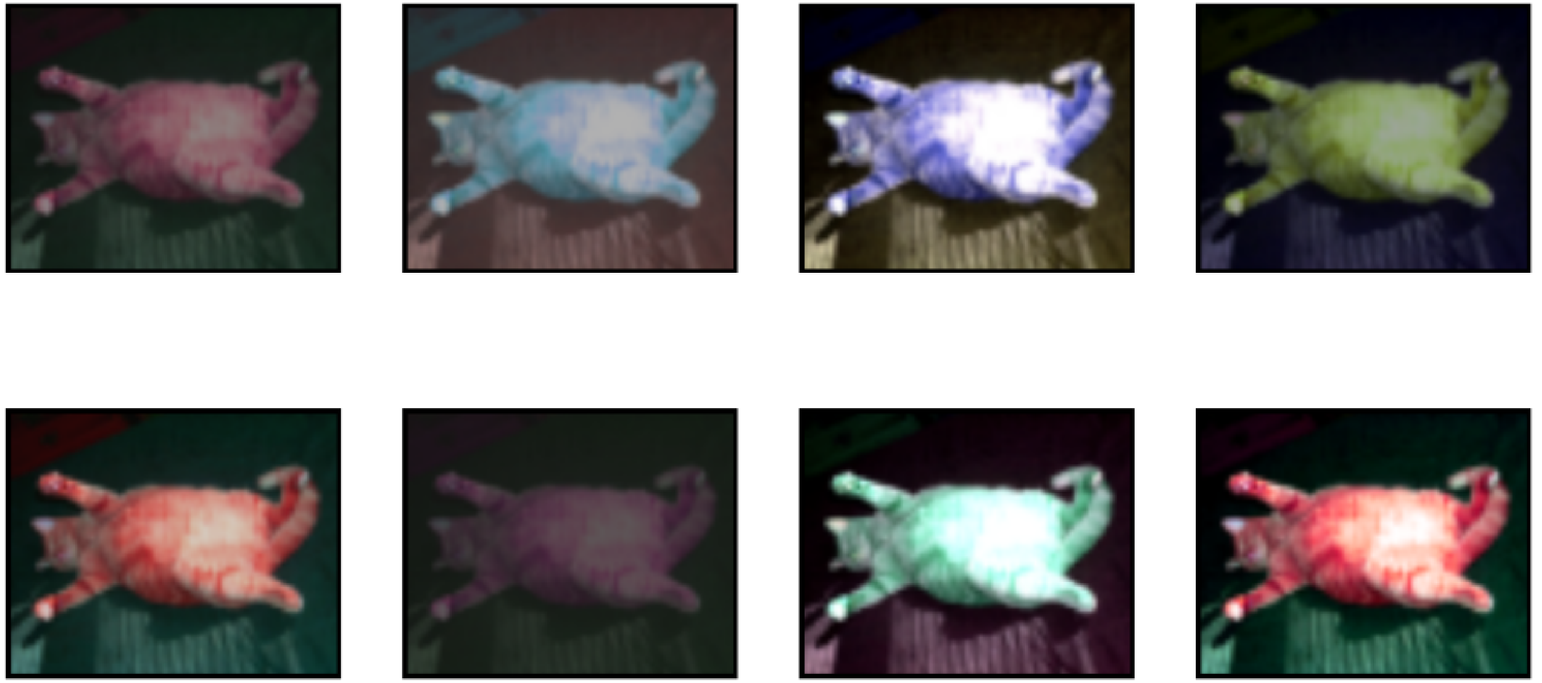
6.1 Image augmentation

- similarly, we can randomly change the hue of the image



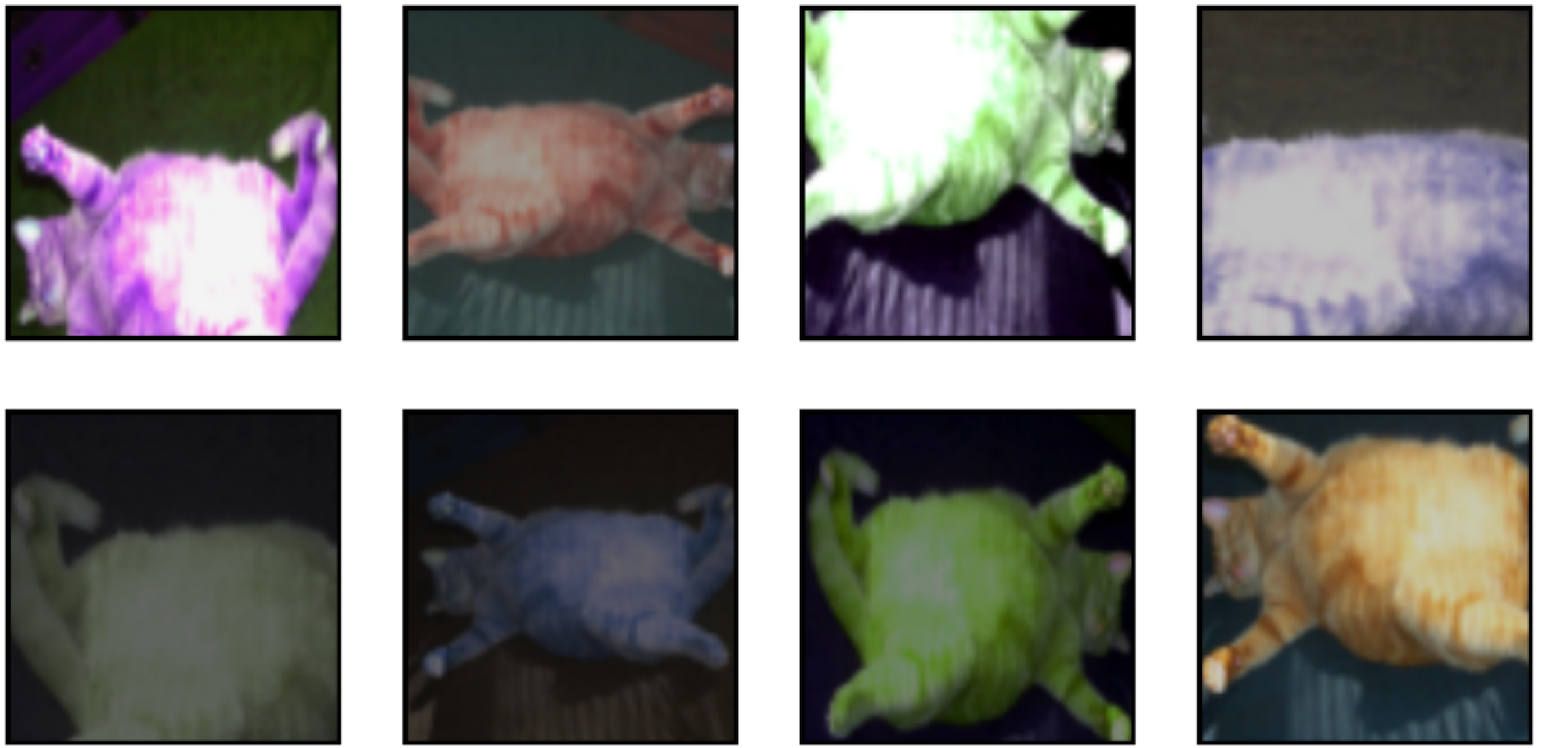
6.1 Image augmentation

- we can also randomly change the brightness, contrast, saturation, and hue of the image at the same time



6.1 Image augmentation

- in practice, we will combine multiple image augmentation methods
- for example, we can combine the different image augmentation methods defined above, and apply them to each image



- in order to obtain definitive results during prediction, we usually only apply image augmentation to training examples, and do not use image augmentation with random operations during prediction